

Cyrus Guided Notes: Tsinghua Automation official site

Tsinghua course pack

2026-05-15

Course source

Official source: <https://www.au.tsinghua.edu.cn/>

Why this course is in the system. Department and research-direction entry point for automation, control science, and intelligent systems.

Learning output. Build an Obsidian field map covering math, signals, control, intelligence, and robotics.

Prerequisite checkpoint

- Linear algebra
- Calculus
- Probability and signals

Formula checkpoint

Do not memorize these first. Read each formula as a sentence: what changes, what is observed, what is optimized, and what output it produces.

$$\dot{\mathbf{x}} = A\mathbf{x} + B\mathbf{u}$$

$$\mathbf{y} = C\mathbf{x} + D\mathbf{u}$$

$$\mathcal{C} = [B \ AB \ \dots \ A^{n-1}B]$$

$$\hat{\mathbf{x}}_{k|k} = \hat{\mathbf{x}}_{k|k-1} + K_k(\mathbf{z}_k - H\hat{\mathbf{x}}_{k|k-1})$$

Visual page

Draw one department map: math -> signals -> control -> estimation -> robotics -> spatial intelligence.

GoodNotes pages

- Page THU-000: automation field map
- Page THU-001: state/input/output vocabulary

Web interaction

A clickable concept graph that opens the correct course pack node.

Obsidian and Notion

Layer	Output
Obsidian	Tsinghua Automation -> field map
Notion	Source status, syllabus status, and expansion blocker.

First study move

Open the official source, copy the prerequisite checkpoint into GoodNotes, then write one formula in your own words before watching or reading more.